APPLIED SCIENCE PRIVATE UNIVERSITY (ASU)

FACULTY OF INFORMATION TECHNOLOGY (FIT)

Homework 2025-1

جامعة العلوم التطبيقية الخاصة
كلية تكنولوجيا المعلومات
تقييم واجب 2025-1

Subject: *Object-Oriented Programming (1) 1301108*

Instructor: *I.Siham Haimour*

Due Date: Nov. 28, 2025

Due Time: 11:59 PM

Student Name: .....

Student ID: .....

□ **CS** □ **SE** □ **CCC** □ **DSAI** □ **Other**

PLEASE NOTE THAT THIS LAB QUIZ IS 2 PAGES AND WITH A TOTAL OF 10 POINTS.

Exam Instructions:

- Use NetBeans IDE to solve the homework question.
- After completing your project, **compress** the entire project folder into a .zip file.
- Upload the compressed .zip file to the **Homework** item in **Edugate**.

Question 1 [10 POINTS]

A. Design a class named **Laptop** that contains the following:

- Three instance data members with private access modifier:
  - **laptopId**: represents the ID of the laptop (integer).
  - **laptopBrand**: represents the brand name of the laptop (String).
  - **laptopRAM**: represents the RAM size in GB (integer).
- One static data member with public access modifier:
  - laptopCounter: keeps track of how many Laptop objects were created (integer).
- A parameterized constructor to initialize the instance variables with the values provided.

  Validation rules must be applied as follows:

  - laptopId must be **positive**; otherwise set to 0.
  - laptopBrand must not be empty; if empty, set to "Unknown".
  - laptopRAM must **be ≥ 4; if value < 4 set to 4**.
- A default constructor that initializes all instance variables to default values (0, "Unknown", 4).
- Each time a Laptop object is created (via any constructor), increment **laptopCounter** by 1.
- Provide setters for all private instance variables. All setter methods should enforce validation rules when applicable (e.g., laptopId cannot be negative, laptopBrand cannot be empty, laptopRAM must be greater than or equal to 4).
- An instance method named **isHighPerformance** that returns true if the RAM size is **16 GB or more**, otherwise false.
- An instance method named **calculateMaintenanceCost()** that returns the annual maintenance cost as follows:
  - If the brand is "**Dell**", cost = $80

- If the brand is "**HP**", cost = $70
- **Otherwise**, cost = $50

- An instance method named **showLaptopDetails()** that prints the values of all data members in the following format:

  Laptop ID: 555

  Laptop Brand: Dell

  Laptop RAM: 16GB

  Total Laptops Created: 12

B. **Write a Demo class that contains the main method.**

1. Ask the user to input the values for laptopId, laptopBrand, and laptopRAM.
2. Create an object of the **Laptop** class named **Lap1** using the **parameterized constructor**.
3. Call the method **setLaptopRAM** for the **Lap1** object to set its ram to **8 GB**.
4. Print the value of the **laptopCounter** static data member.
5. Call the **showLaptopDetails** method using the object **Lap1** .

```java
//Write your code here;
// ======================= Laptop Class =======================
public class Laptop {
    private int laptopId;                      // [0.25 point]
    private String laptopBrand;               // [0.25 point]
    private int laptopRAM;                    // [0.25 point]
    public static int laptopCounter = 0;    // [0.25 point]

    // ================= Default Constructor =================
    public Laptop() {
        setLaptopId(0);
        setLaptopBrand("Unknown");
        setLaptopRAM(4);                      // [1 point]

        laptopCounter++;
    }
    // ================ Parameterized Constructor ================
    public Laptop(int laptopId, String laptopBrand, int laptopRAM) {
        setLaptopId(laptopId);
        setLaptopBrand(laptopBrand);
        setLaptopRAM(laptopRAM);              // [1 point]

        laptopCounter++;
    }
    // ================== Setters with Validation ==================
    public void setLaptopId(int laptopId) {
        if (laptopId > 0) {
            this.laptopId = laptopId;         // [1 point]
        } else {
            this.laptopId = 0;
            System.out.println("Invalid ID. Setting to 0.");
        }
    }

    public void setLaptopBrand(String laptopBrand) {
        if (laptopBrand != null && !laptopBrand.isEmpty()) {
            this.laptopBrand = laptopBrand;
        } else {                               // [1 point]
            this.laptopBrand = "Unknown";
            System.out.println("Invalid brand. Setting to 'Unknown'.");
        }
    }
```

```java
//Write your code here
public void setLaptopRAM(int laptopRAM) {
        if (laptopRAM >= 4) {
            this.laptopRAM = laptopRAM;
        } else {                                // [1 point]
            this.laptopRAM = 4;
            System.out.println("Invalid RAM. Setting to 4GB.");
        }
    }

    // ==================== Instance Methods ====================
    public boolean isHighPerformance() {
        return laptopRAM >= 16;               // [0.5 point]
    }

    public int calculateMaintenanceCost() {  // cost per year
        if (laptopBrand.equalsIgnoreCase("Dell"))
            return 80;                        // [1 point]
        else if (laptopBrand.equalsIgnoreCase("HP"))
            return 70;
        else
            return 50;
    }

    public void showLaptopDetails() {
        System.out.println("Laptop ID: " + laptopId);
        System.out.println("Laptop Brand: " + laptopBrand);  // [0.5 point]
        System.out.println("Laptop RAM: " + laptopRAM + "GB");
        System.out.println("Total Laptops Created: " + laptopCounter);
    }
}
```

```
//Write your code here
// ======================= Democlass =======================
import java.util.Scanner;
public class Demo {
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        System.out.print("Enter Laptop ID: ");
        int id = input.nextInt();

        input.nextLine(); // to avoid scanner bug

        System.out.print("Enter Laptop Brand: ");
        String brand = input.nextLine();

        System.out.print("Enter Laptop RAM: ");
        int ram = input.nextInt();

        Laptop lap1 = new Laptop(id, brand, ram);       // [0.5 point]

        lap1.setLaptopRAM(8);                           // [0.5 point]

        System.out.println("Total Laptops Created: " +
Laptop.laptopCounter);  // [0.5 point]

        lap1.showLaptopDetails();                       // [0.5 point]
    }
}
```

Good luck